

Bubbly Project Summary

Overview:

Bubbly is a market macro-monitoring tool designed to track and quantify potential financial bubbles through a composite index built from key macroeconomic and valuation indicators. It integrates data from multiple online sources such as Yahoo Finance and Shiller's dataset to compute normalized z-scores and generate a 'Bubble Pressure Index' that indicates the current phase of the market cycle.

Goals:

- Detect overvaluation and market bubble build-up phases.
- Combine valuation, liquidity, and volatility indicators into one composite measure.
- Generate interpretable reports and visual dashboards (HTML / PDF).
- Eventually integrate alerts and AI-based adaptive thresholds.

Key Indicators Tracked:

1. **Buffett Ratio** – Market capitalization vs GDP (valuation pressure).
2. **Shiller CAPE** – Cyclically adjusted price-to-earnings ratio (valuation pressure).
3. **M2 YoY Growth** – Liquidity availability in the economy (liquidity trigger).
4. **VIX** – Volatility index as a proxy for market fear (psychological trigger).

Composite Algorithm:

The core of Bubbly is the **Bubble Pressure Index**, a weighted z-score model: $B = 0.4 * Z(\text{Buffett}) + 0.4 * Z(\text{CAPE}) - 0.1 * Z(\text{M2}) - 0.1 * Z(\text{VIX})$ Interpretation:

- Positive values indicate increasing bubble pressure.
- Negative values suggest undervaluation or recessionary environments.
- Thresholds:
 - $B < 1.0 \rightarrow$ Expansion (healthy)
 - $1.0 \leq B < 2.0 \rightarrow$ Euphoria (overheated)
 - $B \geq 2.0 \rightarrow$ Instability (bubble risk).

Data Pipeline:

1. Fetch live and historical data using Yahoo Finance API (via yfinance).
2. Load local Shiller CAPE dataset (.xls) for historical valuation context.
3. Normalize each series to its z-score.
4. Compute weighted composite Bubble Pressure Index.
5. Generate human-readable HTML report (bubbly_report.html).

Future Roadmap:

- Automate scheduled data updates (via cron / systemd).
- Add visualization dashboard (matplotlib + Plotly).
- Implement percentile-based adaptive thresholds.
- Integrate sentiment and credit spread data.
- Deploy on VPS with containerization (Docker) for remote monitoring.